

High-throughput phenotypic traits estimation of faba bean based on machine learning and drone-based multimodal data

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ABSTRACT

Faba bean is a global food legume crop, and it is essential to accurately and timely determine its plant height, above-ground biomass (fresh and dry weight) and yield for enhancing cultivation practices and planning the next planting season. Traditional ground sampling is a time-consuming and labor-intensive approach. However, the utilization of an unmanned aerial vehicle (UAV) as a high-throughput technique offers a promising alternative strategy for estimating crop phenotypic traits. In this study, a two-year experiment was conducted from 2020 to 2022, where UAV-based multimodal data were collected using red–green–blue, multispectral and thermal infrared sensors. The variables derived from these three sensors and their combinations were used to estimate the fresh weight, dry weight and yield of faba bean based on extreme gradient boosting (XGBoost), random forest, multiple linear regression and k-nearest neighbor algorithms. The following findings were obtained: (1) The use of the maximum percentile crop surface model resulted in the highest estimation accuracy for faba bean plant height. (2) Fusion data from multiple sensors increased the estimation accuracy of faba bean fresh weight, dry weight and yield, the coefficient of determination (R^2) improved by 14.22%, 1.45%, and 18.76%, respectively, compared with the best estimation accuracy of a single sensor. (3) The XGBoost algorithm outperformed the other algorithms in estimating fresh weight, dry weight and yield of faba bean. These results demonstrate that multiple sensors and appropriate algorithms can be used to effectively estimate faba bean phenotypic traits and provide valuable insights for agricultural remote sensing research.

1. Introduction

Faba bean (*Vicia faba* L.), a common vegetable and crop, is a delicious, nutritious and essential part of people's daily lives (Temesgen et al., 2015). It was widely cultivated as far back as ancient Egypt, Greece and Rome (Siddiqui et al., 2015). Faba bean is rich in protein, dietary fiber, vitamins and minerals, making it a valuable food source (Garousi et al., 2017). The plant height (PH), above-ground biomass (AGB) and yield are important indicators of the growth, productivity and sustainability of faba bean crops.

Measurements of faba bean PH and AGB can further elucidate plant growth, photosynthetic efficiency, lodging resistance and yield potential (Ji et al., 2022; Wang et al., 2017). The elevated AGB of the faba bean signifies its robust growth and remarkable ecological versatility. These

characteristics significantly bolster nodulation and nitrogen fixation within the plant. Consequently, this leads to an increase in the nitrogen content within the soil, thereby enhancing soil fertility (López-Bellido et al., 2011; Yin et al., 2022). A higher yield of faba bean can provide more food resources for humans, meet local food demands, and improve the income of farmers. However, traditional methods for obtaining PH, AGB and yield rely on manual measurements, which are time-consuming and labor-intensive, making them challenging to apply on a large scale (Fang et al., 2017).

With the continuous advancement of unmanned aerial vehicle (UAV) remote sensing technology and the remote sensing data acquisition has become more efficient and flexible (Bendig et al., 2015; Wang et al., 2019). UAV is a better field observation platform than traditional aerial or satellite remote sensing owing to its cost-effectiveness, high spatial

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resolution and ability to respond quickly to various mission requirements (Yang et al., 2017). Various sensors, such as red–green–blue (RGB), multispectral (MS), hyperspectral, thermal infrared (TIR) and Light Detection and Ranging (LiDAR) sensors, have been designed to be lightweight and deployed on UAVs. These sensors have been successfully used to estimate various phenotypic parameters, such as PH (Volpato and Wright, 2023), AGB (Meng et al., 2023; Randelović et al., 2023), leaf area index (Liu et al., 2021), canopy coverage (CC) (Li et al., 2022) and yield (Ma et al., 2023; Sun et al., 2022). Among these sensors, RGB, MS and TIR are the most promising candidates for widespread use owing to their moderate cost and data processing capabilities (Maimaitijiang et al., 2020). RGB sensors can only provide spectral information in three bands (Liu et al., 2022a), while MS can capture red-edge and near-infrared information, thereby effectively addressing this limitation. Spectral information can capture physiological and biochemical data, and numerous studies have demonstrated the ability of vegetation indices to estimate AGB and yield (Fei et al., 2021a; Ji et al., 2023). However, some studies have shown that spectral saturation may occur as the growth stage progresses, which affects the estimation capability of vegetation indices (Zha et al., 2020; Zhu et al., 2019). Structural information extracted from high-resolution RGB images, such as PH and CC, can capture changes in canopy structure (Liu et al., 2023a). This reflects the health and nutritional status of crops, facilitating the estimation of AGB and yield (Liu et al., 2022b; Liu et al., 2023b). Normalized relative canopy temperature (NRCT) can be used to monitor temporal and spatial changes in soil and crop moisture (Cheng et al., 2023). Crop moisture is closely related to crop growth status, particularly in studies of crop stress (Maimaitijiang et al., 2020). Feature fusion has been applied to several crops, such as potatoes (Liu et al., 2023b) and maize (Shu and Fei, 2022). However, faba bean possess distinct features such as large, broad leaves and a robust stem structure. These morphological traits influence light interception, canopy structure, and biomass distribution, which are crucial for accurate AGB and yield estimation. Hence, the feature fusion should be considered to apply in faba bean, which was limited in previous study. In the extraction of PH, previous researchers mostly used the maximum canopy surface model (CSM) within the area to represent PH. Studies utilizing different percentages of point cloud height to estimate PH are still quite limited.

With the deepening exploration of multiple sensors, many studies have been dissatisfied with the use of individual sensors for estimating crop phenotypic traits. Numerous studies have focused on the fusion of multiple sensor data, which has proven to be effective (Zhu et al., 2021). The fusion of RGB and MS sensors, which is one of the most common fusion strategies, has demonstrated superior performance in estimating maize AGB and soybean yield (Alabi et al., 2022; Hassani et al., 2023). For instance, Cui et al. (2023) estimated the yield of faba bean by fusing RGB and MS sensors, and this fusion method outperformed the use of individual sensors. Feng et al. (2020) used RGB, MS and TIR sensors to estimate cotton yield, and the results showed that the estimation accuracy of multi-feature fusion was higher than that of single feature. Fusion data can combine multiple sensor spectra, structures, and thermal information, which is conducive to the estimation of crop traits. However, studies on the use of TIR sensors for phenotyping food legume crops are relatively few.

ML techniques, such as random forest (RF), support vector machine (SVM), k-nearest neighbor (KNN), multiple linear regression (MLR) and multi-layer perceptron (MLP), are increasingly crucial in phenotype estimation and are widely applied in crop phenotyping. As demonstrated in previous studies, each algorithm has unique characteristics (Maimaitijiang et al., 2017). For example, Selvaraj et al. (2020) estimated cassava AGB using four ML algorithms (RF, KNN, SVM and MLP). The KNN algorithm achieved the highest estimation accuracy, whereas the MLP algorithm achieved the lowest estimation accuracy. In a separate study on estimating forage AGB, the MLP algorithm demonstrated satisfactory levels of estimation accuracy (Franco et al., 2022). Therefore, selecting the appropriate algorithm is crucial for accurate

phenotype estimation. Recently, the extreme gradient boosting (XGBoost) algorithm has attracted the interest of researchers owing to its ability to effectively handle high-dimensional data and complex nonlinear relationships by integrating multiple weak learners and continuously optimizing the estimation performance of the model (Gao et al., 2023; Zhang et al., 2021). Rice yield was successfully estimated with superior accuracy using a hyperspectral sensor and the XGBoost algorithm (Wang et al., 2021). Although the XGBoost algorithm has been promising in recent studies, no studies exist on its application to faba bean.

In this study, we employed a UAV platform with RGB, MS and TIR sensors to collect field images of faba bean over two years. From these images, we extracted structural information, spectral information and temperature data related to the faba bean canopy. These variables were then screened using the Boruta algorithm and used to estimate AGB (fresh weight and dry weight) and yield using XGBoost and three traditional ML methods (RF, KNN and MLR). In addition, the percentile method was applied to estimate faba bean PH. The main objectives of our study are as follows: (1) explore the optimal percentiles for estimating faba bean PH; (2) assess the effectiveness of RGB, MS, TIR and their feature-level fusion data in estimating the fresh weight, dry weight and yield of faba bean; and (3) compare the performance of XGBoost, RF, KNN and MLR in estimating the fresh weight, dry weight and yield of faba bean.

2. Materials and methods

2.1. Experimental site and design

This study was conducted at the experimental base of the Crop Science Research Institute, Chinese Academy of Agricultural Sciences, located in Xinxiang City, Henan Province, China. The coordinates of the experimental base are 113° 45' 40" E, 35° 8' 11" N (Fig. 1).

The experiment was conducted for of two years. In the first year, sowing occurred in November 2020, and harvest occurred in June 2021. A total of 18 varieties (Table S1), each with three replications, were used, resulting in 54 plots. Each plot measured 3.33 m in length and 3 m in width. Subsequently, in the second year, sowing was conducted in November 2021, followed by harvesting in June 2022. A total of 48 varieties (Table S2), replicated three times, were used, amounting to 144 plots. The dimensions of each plot remained consistent at 3.3 m in length and 3 m in width. Prior to sowing, the soil must be meticulously tilled to loosen it, to facilitate the development of plant roots. Concurrently, compound fertilizer should be administered at a rate of 0.6 t/ha. At the time of planting, each plot was composed of 10 rows with 25 faba bean seeds sown per row. Moreover, all seeds were meticulously planted at a depth ranging from 5–8 cm. Throughout both years, standard field management practices were employed.

2.2. Ground data acquisition

During the three critical reproductive stages of the faba bean, namely the flowering stage (April 17, 2021 and April 18, 2022), the podding stage (April 28, 2021 and April 29, 2022) and the grain filling stage (May 8, 2021 and May 10, 2022). PH was determined by measuring 12 randomly selected plants from the ground to the apex of the plant in each plot using a measuring tape, the average value was taken as the ground-measured PH. Three randomly selected plants were placed in sealed bags, and their roots were trimmed before weighing to obtain their fresh weight, which was then converted to t/ha. After measuring their fresh weight, the samples were dried at 80 °C for 96 h until a constant weight was achieved. Finally, the samples were weighed. For yield measurement, all plants were harvested and dehulled for determining the grains weight, and the unit was converted to t/ha.

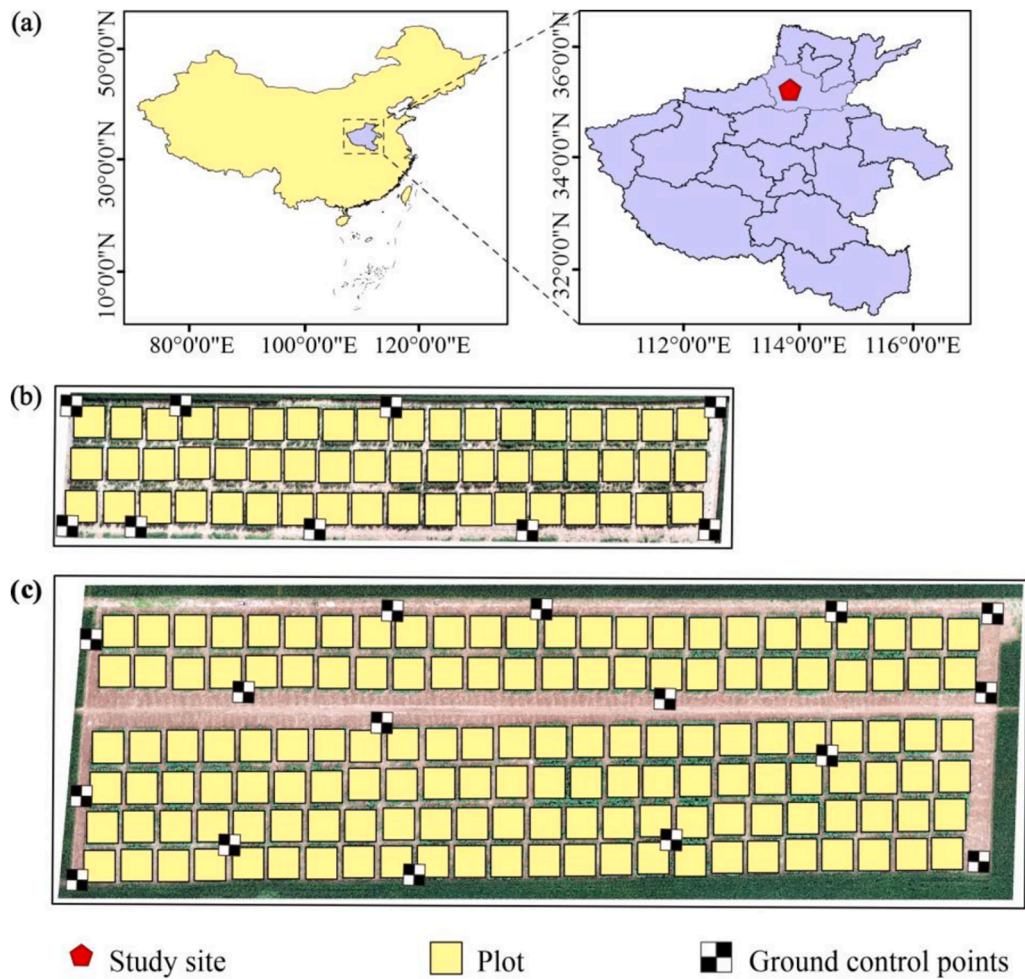


Fig. 1. The layout of the experimental field.

2.3. UAV image acquisition

In this study, RGB images were obtained using a DJI Phantom 4 Pro (SZ DJI Technology Co., Shenzhen, China). The DJI Phantom 4 Pro has a 350 mm wheelbase and a maximum flight speed of 20 m/s, and it captures images at a resolution of 20 million pixels, with the ability to output photos in both JPEG and DNG formats. MS and TIR images were obtained using a DJI M210 (SZ DJI Technology Co., Shenzhen, China) equipped with a RedEdge-MX camera (MicaSense Inc., Seattle, USA) and a Zenmuse XT2 camera (SZ DJI Technology Co., Shenzhen, China). The DJI M210 has a wheelbase of 643 mm, a maximum flight speed of 23 m/s, and a weight of 3.84 kg. The RedEdge-MX camera captures five spectral bands (red, green, blue, red edge and near-infrared), with a resolution of 1.2 million pixels in TIFF format. Moreover, the Zenmuse XT2 camera captures TIR images, with a resolution of 640×512 pixels

in TIFF, JPEG, or R-JPEG format.

The flight altitude was set at 30 m. The RGB images were obtained with a horizontal and vertical overlap rates of 85 %. To obtain images using the MS and TIR sensors, both horizontal and vertical overlap rates were set to 90 %, detailed missions are listed in Table 1. For MS image calibrations, the calibrated reflectance panel was captured both before and after each flight mission. Additionally, a portable thermometer was used to measure the temperature of the blackboard during the flight mission. Ground data collection and UAV image acquisition occurred simultaneously. To ensure high-quality images, flights were scheduled at midday (11:00 AM–1:00 PM). The sensors were equipped with built-in global navigation satellite system (GNSS) modules to obtain positional information from the images. Furthermore, a differential GNSS was used to record the coordinates of the ground control points with millimeter-scale accuracy.

Table 1
Detailed parameters for UAV flights.

Date	Growth stage	Flight time (min)		Flight altitude (m)		Overlap (%)		Ground sample distance (cm)		
		RGB	MS + TIR	RGB	MS + TIR	RGB	MS + TIR	RGB	MS	TIR
20210417	Flowering stage	17	19.6	30	30	85–85	90–90	0.82	2.00	2.38
20210428	Podding stage	17	19.6	30	30	85–85	90–90	0.74	2.05	2.42
20210508	Grain filling stage	17	19.6	30	30	85–85	90–90	0.84	2.11	2.77
20220418	Flowering stage	17	19.6	30	30	85–85	90–90	0.83	2.14	2.70
20220429	Podding stage	17	19.6	30	30	85–85	90–90	0.90	1.98	2.45
20220510	Grain filling stage	17	19.6	30	30	85–85	90–90	0.89	2.02	2.47

Note: RGB: red–green–blue; MS: multispectral; TIR: thermal infrared.

2.4. UAV image processing

The UAV-captured images were preprocessed by applying the Pix4Dmapper (Pix4D, Lausanne, Switzerland). The specific workflow was based on the guidelines provided by Ji et al. (2024).

The workflow of this study is shown in Fig. 2. Notably, calibration is essential for preprocessing both MS and TIR images. For the MS images, a calibrated reflectance panel with a known reflectance of 50 % was utilized to convert the digital number (DN) values to reflectance. For the TIR images, ArcMap 10.5 (Environmental Systems Research Institute, Inc., Redlands, USA) was used to extract the DN values from the blackboard, followed by subsequent calculations to establish the relationship between the actual temperature and DN values. In this study, this relationship was determined as $T = 0.2813 \times \text{DN} - 295.98$.

The processed images were imported into ArcGIS Pro (Environmental Systems Research Institute, Inc., Redlands, USA) and segmented into polygonal regions based on the small areas. Each small area was assigned a unique identifier for further input into ENVI 5.3 (Exelis Visual Information Solutions, Inc., Boulder, USA) to extract area information (Fei et al., 2022). PH was determined by extracting the CSM derived from the difference between the digital surface model and the digital terrain model. Six desired percentiles (PH_{90} , PH_{92} , PH_{95} , PH_{98} , PH_{99} and PH_{max}) of point cloud altitude were extracted using ArcGIS Pro. The faba

bean CC was computed by employing a threshold segmentation method on the RGB image (Castillo-Martínez et al., 2020), with the percentage of plants in the segmented image representing CC, as specified in Equation (1). VIs, which are widely used to estimate crop AGB and yield, were calculated. For the RGB sensor, the VIs were based on DN values, while for the MS sensor, the VIs were based on reflectance, resulting in a total of 35 VIs (NRI: nitrogen reflectance index; RVI: ratio vegetation index; NDVI: normalized difference vegetation index; PNDVI: projected normalized difference vegetation index; MSR: modified simple ratio; DVI: difference vegetation index; MNLI: modified normalized difference leaf index; SAVI: soil adjusted vegetation index; RDVI: renormalized difference vegetation index; GNDVI: green normalized differential vegetation index; NREI: normalized red edge index; MNDI: modified normalized difference index; NNRI: normalized NIR redness index; NDRE: normalized difference red edge index; MEVI: modified enhanced vegetation index; OSAVI: optimized SAVI; GDVI: generalized difference vegetation index; NLI: nonlinear vegetation index; GSAVI: green soil adjusted vegetation index; MCARI1: modified chlorophyll absorption in reflectance index; GRDVI: green red difference vegetation index; GOSAVI: green optimal soil adjusted vegetation index; EXGR: excess green minus excess red index; EXG: excess green index; RGBVI: red green blue vegetation index; GLI: green leaf index; CIVE: color index of vegetation; GRVI: green red vegetation index; NDI: normalized

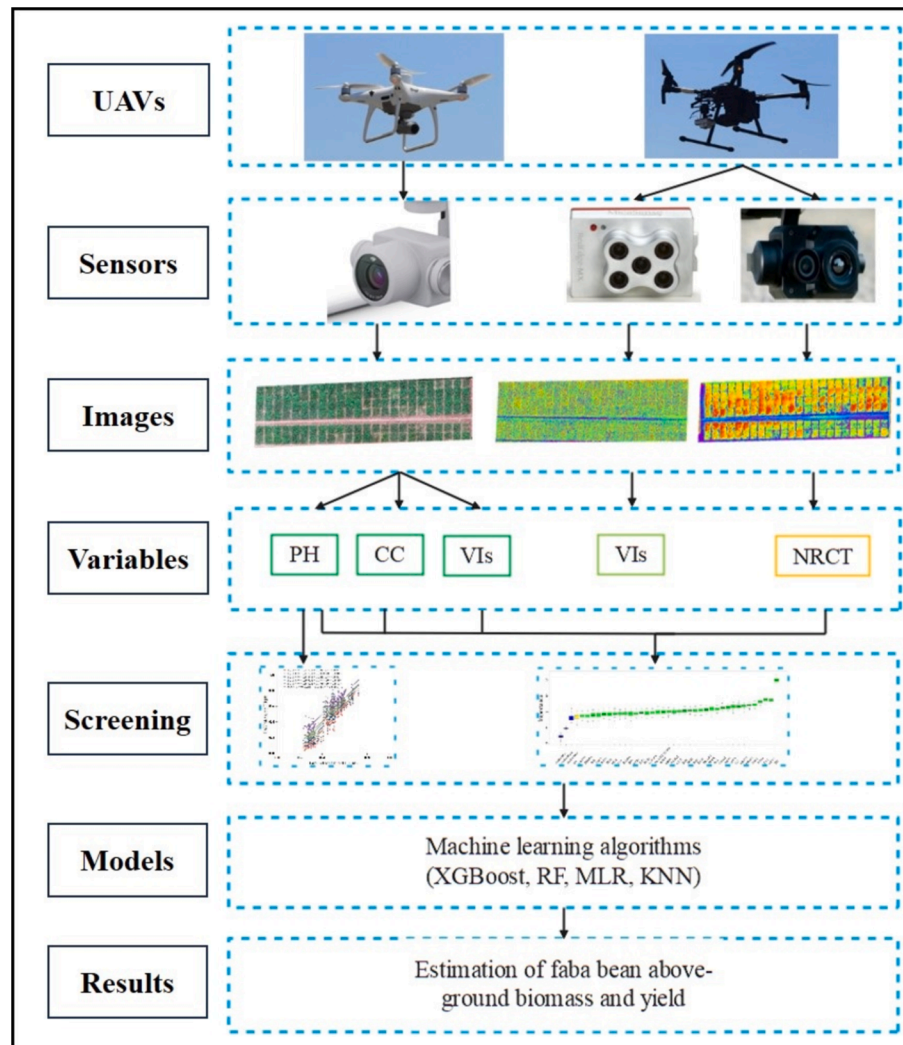


Fig. 2. Diagram showing workflow of observation platforms, sensors, features and models. UAV: unmanned aerial vehicle; PH: plant height; CC: canopy coverage; VIs: vegetation indices; NRCT: normalized relative canopy temperature; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor.

difference index; VARI: visible atmospherically resistant index; GLI2: green leaf index 2; WI: Woebbecke index; MGRVI: modified green red vegetation index; EXR: excess red index). The formulas for calculating the VIs are presented in Table S3. The TIR sensor was used for compute NRCT, as defined in Equation (2).

$$\text{Canopy coverage} = P_{\text{canopy}} / P_{\text{total}} \quad (1)$$

where P_{canopy} denotes the pixels number of faba bean canopy; P_{total} denotes the total pixels number of sampled plot.

$$\text{NRCT} = (T_i - T_{\min}) / (T_{\max} - T_{\min}) \quad (2)$$

where T_i denotes the temperature of pixel i ; $T_{\min}$ denotes the minimum temperature in the whole field, and $T_{\max}$ denotes the maximum temperature in the whole field.

2.5. ML algorithms

In this study, to ensure the robustness of estimating faba bean phenotypic traits, the Boruta algorithm was adopted to remove sensitive variables to construct the dataset. Four algorithms, namely XGBoost, RF, MLR and KNN, were used to investigate fresh weight, dry weight and yield of faba bean. This process was implemented using the R software packages.

The Boruta algorithm was developed as a wrapper for the RF classification algorithm to identify all relevant variables (Shu et al., 2021), thereby creating a novel feature selection algorithm. Throughout the selection process, an iterative method was used to exclude features with a lower correlation with the random probes, resulting in the retention of the selected features. This added element of randomness allows for a better understanding of the attributes with true importance.

The XGBoost algorithm, developed by Chen from the University of Washington (Chen and Guestrin, 2016), is an improved ensemble learning algorithm that is based on the gradient boosting tree algorithm. It follows the principle of adding trees and performing feature splits to enhance its predictive capabilities. Each additional tree learns a new function to better fit the previous prediction. By mapping the features of the samples, each decision tree assigns the samples to specific leaf nodes. The weight score assigned to each leaf node represents the predicted value of the sample. Summing the scores from all trees yields the final prediction for each sample.

RF, proposed by Leo. Breimanl, (2001) as an improvement on bagging, comprises numerous decision trees. It is suitable for both classification and regression tasks. In regression, the prediction result is calculated as the average of all decision trees. This voting mechanism significantly improves the prediction accuracy of the RF model and effectively mitigates common issues, such as overfitting in regression algorithms (Cutler et al., 2007). RF handles nonlinear multiple combination regression problems, making it highly applicable to practical scenarios.

In regression analysis, MLR involves modeling the relationship between multiple independent variables and one or more dependent variables. In this study, the dependent variables were predicted using an optimal combination of multiple independent variables. This approach proves to be more effective and realistic compared with relying solely on a single independent variable.

The KNN algorithm, initially proposed by Cover and Hart (1953), is a well-established algorithm that is easy to understand, has low computational requirements, and is widely applicable. In a regression task, the KNN finds the k -nearest training samples to a given test sample using a distance measure. The label value of the sample is then calculated as the arithmetic average of the label values of the k -nearest samples. By performing distance calculations, typically employing the Euclidean or Manhattan distance measures, KNN overcomes issues related to object matching and serves various practical applications (Muslikhin et al., 2020).

2.6. Estimation accuracy evaluation

In order to fairly and completely compare different algorithms in estimating faba bean AGB and yield, we randomly selected 80 % of the samples as training datasets, and the remaining 20 % were selected as testing datasets. A five-fold cross-validation was applied to test the accuracy of the estimations, thereby ensuring that all samples were used independently for the validation.

In this study, R^2 , RMSE, and normalized root mean square error (NRMSE) were used to evaluate the robustness of the model for estimating the fresh weight, dry weight and yield of faba bean (Yu et al., 2020). The calculation formulas are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (X_i - \hat{X}_i)^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (X_i - \hat{X}_i)^2}{n}} \quad (4)$$

$$\text{NRMSE} = \text{RMSE} / \bar{X} \quad (5)$$

where n denotes the total number of samples; X_i and $\hat{X}_i$ denote the measured and estimated PH, AGB or yield of samples, respectively; and $\bar{X}$ denotes the mean of the measured PH, AGB or yield.

3. Results

3.1. Statistical description of faba bean phenotypic traits

The ground measurements of PH, fresh weight, dry weight and yield in two years are presented in Table 2. In 2021, the PH values range from 0.24 – 0.73 m, fresh weight exhibits a range of 5.9—39.2 t/ha, dry weight spans from 0.71 – 5.48 t/ha, and yield varies from 1.5 – 4.6 t/ha. In 2022, the PH values range from 0.16 – 0.85 m, fresh weight exhibits a range of 1.4—45.3 t/ha, dry weight exhibits a range of 0.08—10.3 t/ha, and yield exhibits a range of 0.3—5.9 t/ha. The large data span also provides the basis for the robustness of the model. From Table 2, it can be observed that the average PH and dry weight of faba bean in 2022 were higher than those of 2021, with an increase of 0.07 m and 0.49 t/ha, respectively. However, the average fresh weight and yield were slightly lower in 2021, with decreases of 0.8 and 0.17 t/ha, respectively.

3.2. Estimation of faba bean PH

To enhance the estimation of faba bean PH, various percentiles (PH_{90} , PH_{92} , PH_{95} , PH_{98} , PH_{99} and $PH_{\max}$) of the CSM were adopted to extract PH at the plot scale. Subsequently, the extracted PH values were fitted with the ground measurements. These findings consistently demonstrated a stable level of estimation accuracy when various percentiles were employed (Table 3). Notably, in 2021, the best fit was achieved at PH_{90} with an R^2 value of 0.8699, while $PH_{\max}$ had the smallest error with an RMSE of 0.0858 m. In 2022, the $PH_{\max}$ percentile exhibited the best fit ($R^2 = 0.8871$) and the smallest error for PH estimation. Fig. 3 shows the best estimates visually. In 2021, the highest R^2 value was achieved at PH_{90} , which also led to some overestimation of PH. However, this had a limited impact on the overall estimation accuracy. Therefore, the $PH_{\max}$ was selected as a variable of structure information for estimating the fresh weight, dry weight and yield of faba bean.

3.3. Estimation of faba bean fresh weight

The Boruta algorithm was used to remove out the sensitive variables related to the fresh weight of the faba bean in this study. The screening

Table 2
Descriptive statistics for ground measured parameters.

Parameters	Date	No. of samples	Min	Max	Mean	SD	CV
PH (m)	20210417	54	0.24	0.41	0.31	0.04	12 %
	20220418	144	0.16	0.53	0.39	0.05	13 %
	20210428	54	0.41	0.58	0.49	0.05	9 %
	20220429	144	0.32	0.76	0.56	0.08	14 %
	20210508	54	0.48	0.73	0.57	0.06	10 %
	20220510	144	0.44	0.85	0.65	0.08	12 %
F (t/ha)	20210417	54	5.93	20.03	12.76	3.54	28 %
	20220418	144	1.41	17.47	9.41	3.53	37 %
	20210428	54	10.78	28.34	19.51	4.29	22 %
	20220429	144	4.26	33.5	21.75	6.31	29 %
	20210508	54	20.25	39.24	29.89	5.11	17 %
	20220510	144	7.64	45.33	28.6	8.61	30 %
D (t/ha)	20210417	54	0.71	3.71	2.14	0.67	31 %
	20220418	144	0.08	3.16	1.48	0.63	42 %
	20210428	54	1.22	4.98	3.02	0.84	28 %
	20220429	144	0.69	5.37	2.97	0.86	29 %
	20210508	54	2.01	5.48	3.9	1.01	26 %
	20220510	144	1.66	10.33	5.24	1.5	29 %
Yield (t/ha)	2021	54	1.5	4.85	3.07	0.9	29 %
	2022	144	0.3	5.9	2.9	1.24	43 %

Note:SD: standard deviation; CV: coefficient of variation; PH: plant height; F: fresh weight; D: dry weight.

Table 3
Comparison of extracted plant height from different percentiles with ground measured values.

Date	Percentile	Fitting equation	R ²	RMSE (m)
2021	PH ₉₀	y = 0.5342x + 0.251	0.8699	0.1294
2021	PH ₉₂	y = 0.5372x + 0.2426	0.8684	0.1215
2021	PH ₉₅	y = 0.5464x + 0.225	0.8647	0.1084
2021	PH ₉₈	y = 0.5653x + 0.1948	0.8586	0.0977
2021	PH ₉₉	y = 0.5783x + 0.1756	0.8577	0.0972
2021	PH _{max}	y = 0.6325x + 0.094	0.8586	0.0858
2022	PH ₉₀	y = 0.6873x + 0.2309	0.8817	0.1164
2022	PH ₉₂	y = 0.6943x + 0.223	0.8829	0.1099
2022	PH ₉₅	y = 0.7067x + 0.2079	0.8843	0.0981
2022	PH ₉₈	y = 0.725x + 0.1841	0.8854	0.0810
2022	PH ₉₉	y = 0.7383x + 0.1688	0.8854	0.0722
2022	PH _{max}	y = 0.7711x + 0.1329	0.8871	0.0572

Note: PH₉₀, PH₉₂, PH₉₅, PH₉₈, PH₉₉, PH_{max} represent the 90, 92, 95, 98, 99, 100 percentile estimates of faba bean plant height value.

results for the three growth stages are illustrated in Fig. 4. At the flowering stage, all variables were deemed important except for WI. In the subsequent podding stage, all variables were recognized as important. However, at the grain filling stage, four variables (WI, NRCT, MNLI and RDVI) were deemed uncertain, while the remaining variables were still considered important. Although several variables were uncertain, they were all within the acceptable range. Therefore, all variables were

included in the dataset to estimate the fresh weight throughout the three growth stages.

This study combined three sensors and their fusion data with four ML algorithms (XGBoost, RF, MLR and KNN) to estimate the faba bean fresh weight at three growth stages. The detailed estimation results are presented in Table 4. Fig. 5 shows the difference in the estimation of faba bean fresh weight between different sensors and different algorithms. As depicted in Fig. 5A, when using an individual sensor to estimate fresh weight, the RGB sensor achieved the highest estimation accuracy ($R^2 = 0.27\text{--}0.66$), while the TIR sensor achieved the lowest estimation accuracy ($R^2 = 0.10\text{--}0.38$). When using fusion data to estimate the faba bean fresh weight, all estimation accuracies improved to some extent. Among them, RGB + MS + TIR (RMT) exhibited the highest estimation accuracy, with R^2 values reaching 0.75, which represents a 14.22 % improvement over the best estimation accuracy of a single sensor, followed by RGB + MS (RM), with R^2 values ranging from 0.36 to 0.75. This suggests that the variables from the RGB sensor contributed the most to faba bean fresh weight estimation. Furthermore, the XGBoost algorithm outperformed the other three algorithms in estimating faba bean fresh weight (Fig. 5B), with the highest R^2 value of 0.75, followed by the RF and MLR algorithms, with optimal R^2 values of 0.67 and 0.59, respectively. The KNN achieved the lowest estimation accuracy. Therefore, the XGBoost algorithm was the optimal choice for estimating the fresh weight of the faba bean in this study.

3.4. Estimation of faba bean dry weight

Fig. 6 illustrates the variable selection results for dry weight using the Boruta algorithm. At the flowering and podding stages, the NRCT variable was determined to be unimportant, while the remaining 37 variables were included as input variables. At the grain filling stage, in addition to the eight unimportant variables (ExG, WI, CIVE, RGBVI, PH, GLI, ExGR and NRCT), other important and uncertain variables (GRDVI, OSAVI, EXR, NDI, CC, NLI, GOSAVI, NRI, GRVI, MCARI1/OSAVI, MGRVI, MEVI, RVI, GNDVI, MSR, NDVI, MCARI1, MNDI, NNRI, VARI, GLI2, NREI, NDRE, PNDVI, GDVI, DVI, MNLI, RDVI, SAVI and GSAVI) were used to construct the dataset to estimate the dry weight of faba bean.

Due to the variable NRCT was deemed unimportant for estimating faba bean dry weight, the TIR sensor was not utilized in the estimation process. Table 5 presents the estimation results obtained using four ML algorithms based on two sensors and their fusion data. Fig. 7 visually shows the difference in estimating faba bean dry weight. When using a single sensor, the RGB sensor consistently outperformed the MS sensor across all four algorithms (XGBoost, RF, MLR and KNN), with an average R^2 increase of 0.04, 0.10, 0.10 and 0.12, respectively. The use of RM fusion data with XGBoost and RF algorithms obtained higher estimation accuracy than using a single sensor, with average R^2 values of 0.59 and 0.55, respectively. However, the use of the MLR and KNN algorithms with fusion data obtained lower estimation accuracy than the RGB sensor.

The XGBoost algorithm achieved maximum estimation accuracy values, with R^2 reaching as high as 0.72, 0.57 and 0.73 at three growth stages when the fusion data were used. The average estimation accuracy of XGBoost surpassed those of the other three algorithms, except when the RGB sensor was used. In summary, the XGBoost algorithm outperformed the other three algorithms (MLR, RF and KNN) in estimating the faba bean dry weight.

3.5. Estimation of faba bean yield

As presented in Fig. 8, 25 important variables (NRI, RVI, NDVI, PNDVI, MSR, DVI, MNLI, SAVI, RDVI, GNDVI, NREI, MNDI, NNRI, NDRE, CC, MEVI, OSAVI, NRCT, GDVI, NLI, GSAVI, MCARI1/OSAVI, MCARI1, GRDVI and GOSAVI) and 5 uncertain variables (EXGR, EXG, RGBVI, GLI and CIVE) were screened out at the flowering stage. At the

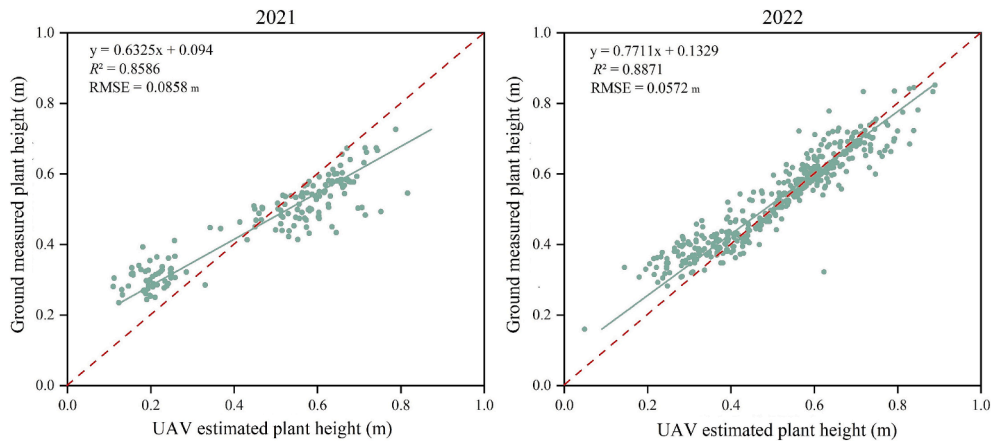


Fig. 3. Scatter plot of ground measured and best estimated plant height (PH_{max}).

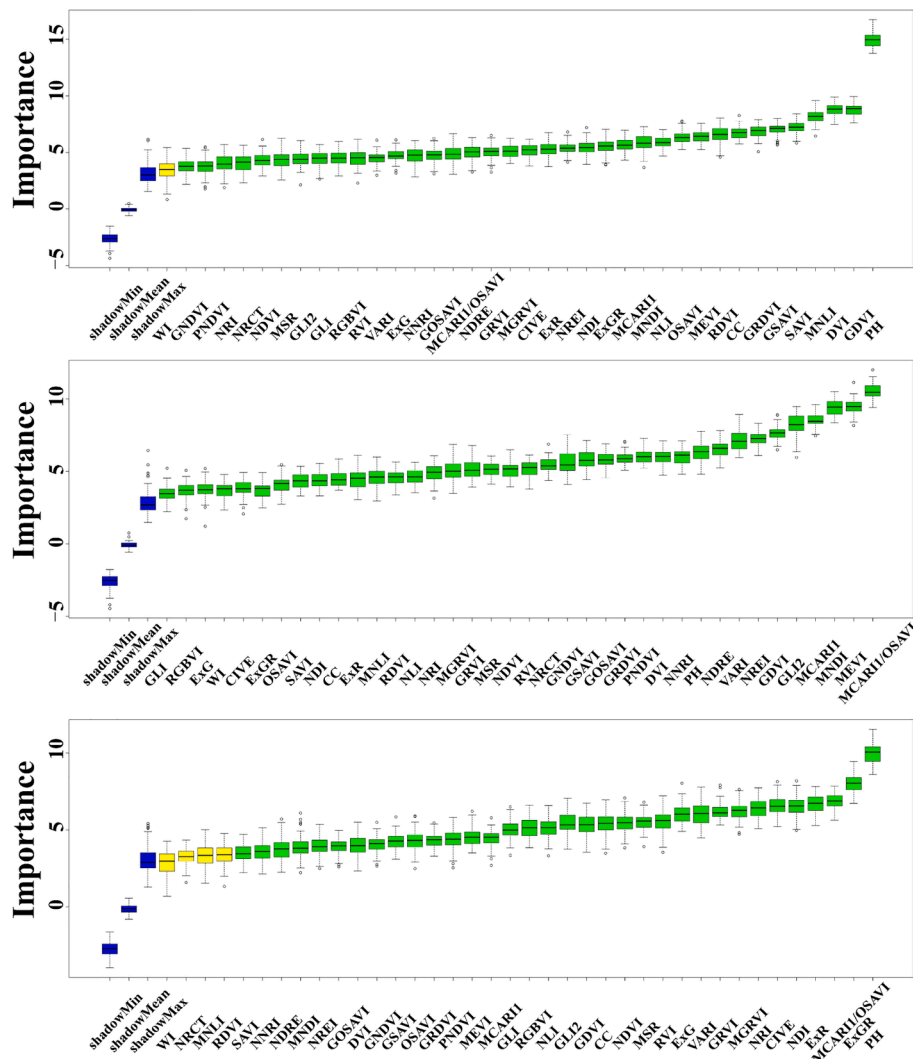


Fig. 4. Screening process of sensitive variables for fresh weight. The blue box charts depicted the minimum, average and maximum Z scores for the shadow attribute, while the red, yellow and green box charts sequentially displayed the Z scores for the unimportant, uncertain and important variables. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

podding and grain filling stages, all variables were determined to be important and were used to estimate faba bean yield.

In this study, three sensors and their combinations were utilized at three growth stages to estimate faba bean yield using four ML algorithms

to identify the optimal combination for faba bean yield. Detailed results are presented in Table 6. The difference of sensor estimation accuracy is shown in Fig. 9A. When using a single sensor, the MS sensor ($R^2 = 0.54$, $RMSE = 0.80t/ha$ and $NRMSE = 27.25\%$) exhibited the highest average

Table 4

The details of the estimation results for fresh weight.

		XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN
R	R^2	0.66	0.65	0.57	0.27	0.61	0.56	0.61	0.55	0.52	0.51	0.48	0.30
	RMSE (t/ha)	2.17	2.30	2.60	3.27	3.62	3.98	3.71	4.03	5.98	5.60	5.84	6.13
	NRMSE	21.16 %	22.24 %	25.22 %	31.70 %	17.05 %	18.80 %	17.57 %	19.04 %	20.56 %	19.02 %	20.06 %	22.69 %
M	R^2	0.48	0.54	0.50	0.38	0.51	0.63	0.60	0.42	0.48	0.45	0.47	0.24
	RMSE (t/ha)	2.82	2.65	2.80	3.02	4.19	3.63	3.87	4.49	5.68	5.80	5.78	6.86
	NRMSE	27.72 %	25.64 %	27.16 %	29.20 %	19.70 %	17.17 %	18.31 %	21.22 %	19.58 %	20.04 %	19.95 %	23.70 %
T	R^2	0.21	0.10	0.14	0.18	0.38	0.27	0.18	0.29	0.23	0.23	0.25	0.23
	RMSE (t/ha)	3.33	3.94	3.56	3.46	4.94	5.26	5.35	5.10	6.60	7.30	6.93	7.01
	NRMSE	32.45 %	38.20 %	34.49 %	33.49 %	23.42 %	24.86 %	25.31 %	24.14 %	22.32 %	25.20 %	23.95 %	24.20 %
RM	R^2	0.75	0.66	0.59	0.53	0.64	0.64	0.65	0.58	0.53	0.48	0.50	0.36
	RMSE (t/ha)	1.99	2.25	2.47	2.61	3.91	3.60	3.58	3.88	5.28	5.66	5.53	6.33
	NRMSE	19.56 %	21.81 %	23.92 %	25.24 %	18.58 %	17.01 %	16.96 %	18.35 %	18.13 %	19.54 %	19.12 %	21.88 %
RT	R^2	0.67	0.66	0.59	0.29	0.59	0.57	0.62	0.45	0.53	0.49	0.48	0.39
	RMSE (t/ha)	2.22	2.25	2.48	3.22	3.72	3.96	3.68	4.47	5.95	5.51	5.81	6.57
	NRMSE	21.27 %	21.76 %	24.06 %	31.20 %	17.53 %	18.73 %	17.42 %	21.13 %	18.83 %	19.33 %	20.18 %	21.17 %
MT	R^2	0.61	0.55	0.51	0.28	0.51	0.62	0.59	0.45	0.41	0.45	0.47	0.35
	RMSE (t/ha)	2.40	2.59	2.77	3.23	3.84	3.67	3.90	4.40	6.38	5.79	5.80	6.31
	NRMSE	23.33 %	25.08 %	26.79 %	31.31 %	17.85 %	17.36 %	18.43 %	20.80 %	21.95 %	20.01 %	20.04 %	21.80 %
RMT	R^2	0.75	0.67	0.57	0.63	0.70	0.64	0.65	0.58	0.56	0.48	0.50	0.45
	RMSE (t/ha)	1.92	2.21	2.60	2.36	3.38	3.60	3.60	3.86	5.45	5.61	5.51	5.79
	NRMSE	18.60 %	21.37 %	25.22 %	22.83 %	16.17 %	17.04 %	17.05 %	18.27 %	17.19 %	19.38 %	19.03 %	20.00 %

Note: R: red–green–blue (RGB); M: multispectral (MS); T: thermal infrared (TIR); RM: RGB + MS; RT: RGB + TIR; MT: MS + TIR; RMT: RGB + MS + TIR; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor.

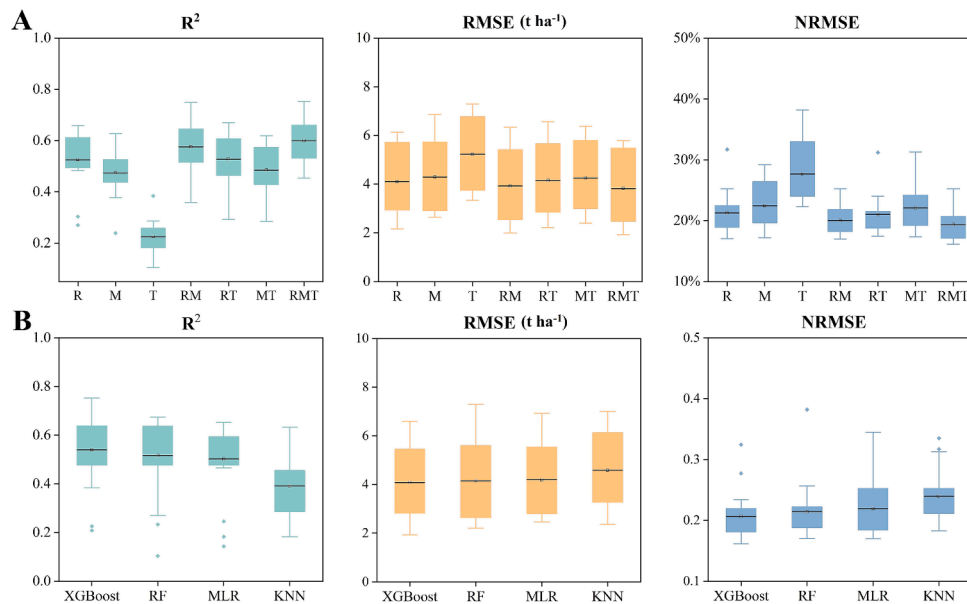


Fig. 5. The statistical distribution of the estimation accuracy for fresh weight. A: comparison the estimation accuracy of different sensors; B: comparison the estimation accuracy of different models. XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor; R: red–green–blue (RGB); M: multispectral (MS); T: thermal infrared (TIR); RM: RGB + MS; RT: RGB + TIR; MT: MS + TIR; RMT: RGB + MS + TIR. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

estimation accuracy, followed by the RGB sensor ($R^2 = 0.51$, RMSE = 0.82t/ha and NRMSE = 27.89 %), while the TIR sensor exhibited the lowest performance ($R^2 = 0.36$, RMSE = 0.95t/ha and NRMSE = 32.34 %). However, the RGB sensor exhibited the highest estimation accuracy ($R^2 = 0.67$, RMSE = 0.68t/ha and NRMSE = 22.96 %). When multiple sensors were used, RMT exhibited a higher average estimation accuracy ($R^2 = 0.61$, RMSE = 0.75t/ha and NRMSE = 25.33 %) than the dual-sensors, and it exhibited the highest estimation accuracy ($R^2 = 0.79$, RMSE = 0.62t/ha and NRMSE = 21.55 %), which represents a 18.76 % improvement over the highest R^2 value achieved by applying a single sensor.

Fig. 9B shows the difference in estimating faba bean yield by different algorithms and different growth stages. The XGBoost algorithm exhibited higher average estimation accuracy ($R^2 = 0.58$, RMSE =

0.77t/ha and NRMSE = 26.47 %) than the other three algorithms (RF, MLR and KNN) and obtained the highest estimation accuracy at all three stages (flowering, podding and grain filling stage), with R^2 values of 0.61, 0.65 and 0.79, respectively. The MLR algorithm ranked second, with R^2 values of 0.58, 0.63 and 0.70 at the three growth stages. The KNN algorithm exhibited the lowest estimation accuracy. The estimation accuracy of all four ML algorithms improved as the growth stages progressed, and the R^2 for the grain filling stage was 0.18 higher than that of flowering stage. In conclusion, the combination of RMT fusion data and XGBoost algorithm was the optimal choice for estimating faba bean yield in this study.

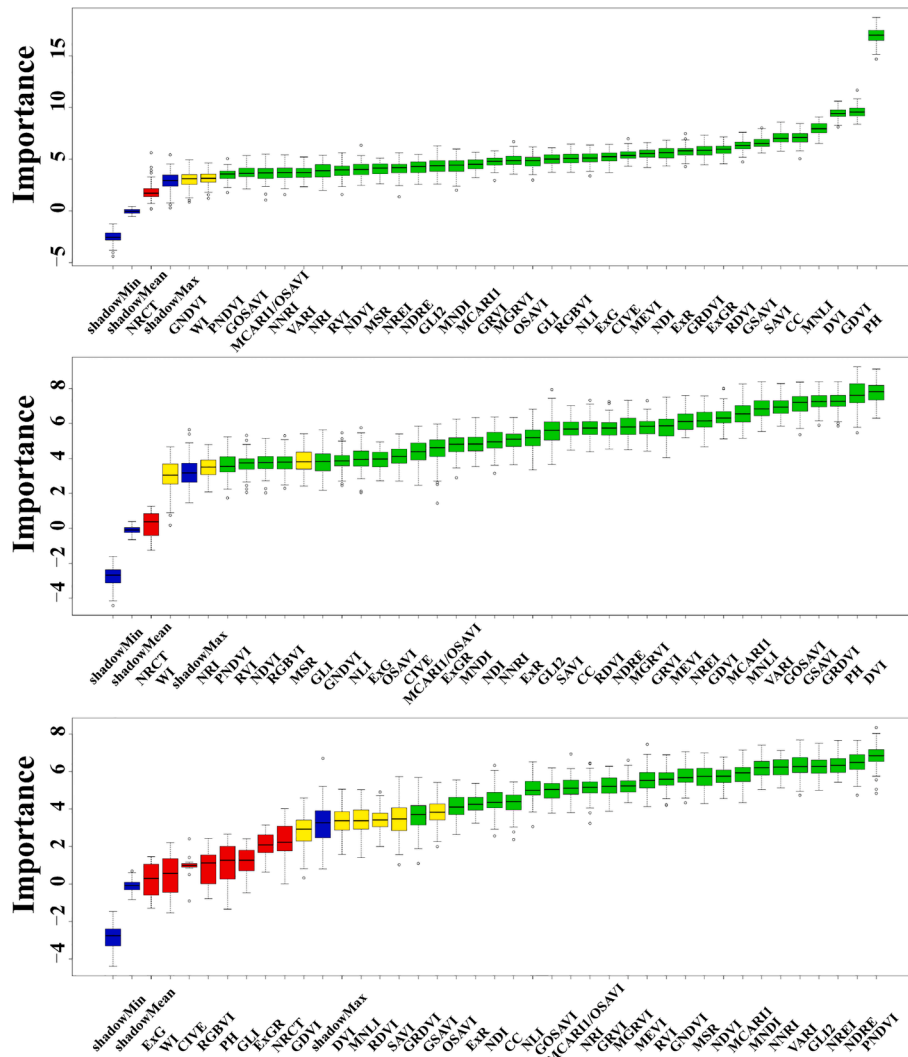


Fig. 6. Screening process of sensitive variables for dry weight. The blue box charts depicted the minimum, average and maximum Z scores for the shadow attribute, while the red, yellow and green box charts sequentially displayed the Z scores for the unimportant, uncertain and important variables. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

The details of the estimation results for dry weight.

		Flowering stage				Podding stage				Grain filling stage			
		XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN
R	R^2	0.72	0.70	0.65	0.66	0.52	0.56	0.58	0.51	0.42	0.40	0.48	0.42
	RMSE (t/ha)	0.38	0.39	0.42	0.41	0.59	0.61	0.58	0.64	1.22	1.18	1.10	1.16
	NRMSE	22.91 %	23.67 %	25.58 %	24.87 %	21.46 %	22.23 %	21.26 %	23.18 %	24.76 %	24.27 %	22.50 %	23.73 %
M	R^2	0.58	0.52	0.49	0.40	0.48	0.47	0.50	0.41	0.49	0.35	0.41	0.41
	RMSE (t/ha)	0.48	0.49	0.53	0.54	0.61	0.65	0.65	0.71	1.01	1.25	1.21	1.16
	NRMSE	29.00 %	29.60 %	31.85 %	32.84 %	22.29 %	23.81 %	23.69 %	25.66 %	20.66 %	25.62 %	24.82 %	23.91 %
RM	R^2	0.74	0.71	0.62	0.53	0.55	0.51	0.54	0.41	0.50	0.42	0.39	0.41
	RMSE (t/ha)	0.37	0.38	0.46	0.48	0.57	0.63	0.62	0.70	0.99	1.16	1.27	1.17
	NRMSE	22.85 %	23.02 %	27.66 %	28.91 %	20.80 %	22.81 %	22.65 %	25.60 %	20.29 %	23.79 %	26.10 %	24.04 %

Note: R: red–green–blue (RGB); M: multispectral (MS); RM: RGB + MS; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor.

4. Discussion

4.1. Analysis of pH estimation

In previous studies on the PH of faba bean, PH estimation was primarily based on the maximum statistical value (Stubbs et al., 2023, Yan

et al., 2023). Currently, no studies exist on the use of percentile values to estimate faba bean PH. Previous studies have demonstrated that different percentile heights are applicable to various crops and sensor data. Zang et al. (2023) discovered that PH_{99} was the most suitable percentile for estimating wheat PH. However, this study achieved more accurate estimation results using estimation data derived from PH_{max} .

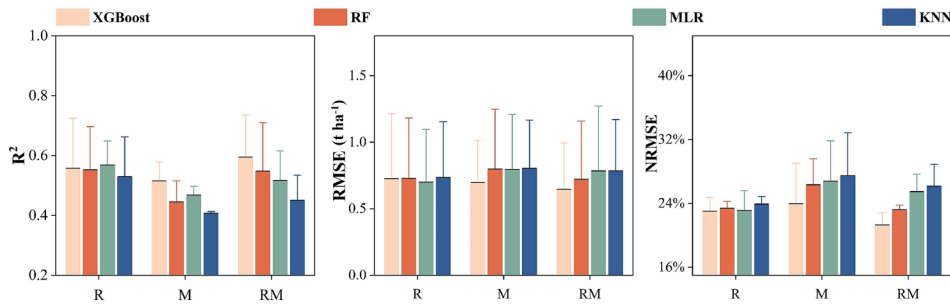


Fig. 7. The estimation accuracy of dry weight with different sensors and models. R: red–green–blue (RGB); M: multispectral (MS); RM: RGB + MS; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

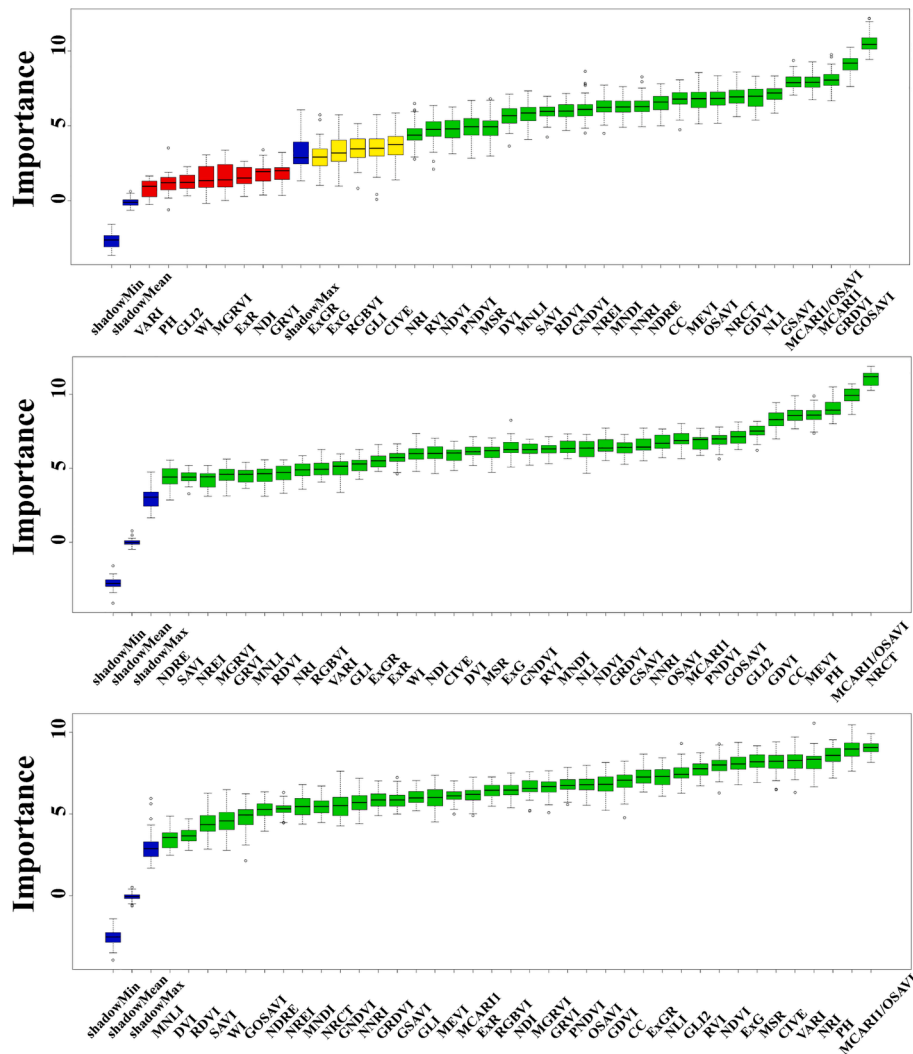


Fig. 8. Screening process of sensitive variables for yield. The blue box charts depicted the minimum, average and maximum Z scores for the shadow attribute, while the red, yellow and green box charts sequentially displayed the Z scores for the unimportant, uncertain and important variables. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Employing percentile height for PH estimation can partially mitigate the influence of outliers on pixel values (Han et al., 2019).

4.2. Performance analysis of faba bean AGB estimation

VIs extracted from UAV imagery have been widely used to estimate the fresh and dry weights of crops (Randelović et al., 2023). In this study,

we employed the Boruta algorithm to select sensitive VIs and subsequently estimated the AGB of faba bean using three sensors and their combinations. It is noteworthy that NRCT is closely related to crop transpiration and water status. It is an important feature in estimating fresh weight but has been identified as an unimportant feature in estimating dry weight. This may be because crops with a high water content usually maintain lower canopy temperatures through transpiration, and

Table 6

The details of the estimation results for yield.

		Flowering stage				Podding stage				Grain filling stage			
		XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN	XGBoost	RF	MLR	KNN
R	R^2	0.50	0.33	0.37	0.35	0.59	0.55	0.60	0.50	0.60	0.67	0.61	0.45
	RMSE (t/ha)	0.80	0.96	0.93	0.95	0.78	0.78	0.74	0.84	0.72	0.68	0.79	0.86
	NRMSE	28.69 %	32.57 %	31.54 %	32.16 %	26.53 %	26.54 %	25.04 %	28.36 %	24.38 %	22.96 %	26.76 %	29.17 %
M	R^2	0.52	0.47	0.58	0.42	0.62	0.53	0.56	0.45	0.64	0.63	0.62	0.40
	RMSE (t/ha)	0.82	0.85	0.77	0.90	0.69	0.79	0.78	0.87	0.77	0.71	0.76	0.90
	NRMSE	27.46 %	28.78 %	26.24 %	30.55 %	24.74 %	26.96 %	26.75 %	29.47 %	25.88 %	24.02 %	25.75 %	30.42 %
T	R^2	0.35	0.30	0.32	0.31	0.43	0.32	0.39	0.42	0.47	0.30	0.37	0.36
	RMSE (t/ha)	0.97	1.03	0.96	0.99	0.93	0.99	0.91	0.89	0.84	1.03	0.93	0.95
	NRMSE	33.38 %	34.97 %	32.75 %	33.57 %	31.69 %	33.46 %	30.85 %	30.17 %	28.47 %	34.87 %	31.72 %	32.17 %
RM	R^2	0.56	0.49	0.56	0.41	0.65	0.60	0.63	0.55	0.70	0.67	0.70	0.65
	RMSE (t/ha)	0.79	0.83	0.80	0.90	0.65	0.74	0.71	0.79	0.64	0.68	0.65	0.69
	NRMSE	26.07 %	28.31 %	27.03 %	30.65 %	23.11 %	25.13 %	23.99 %	26.69 %	21.93 %	23.10 %	22.06 %	23.34 %
RT	R^2	0.43	0.39	0.38	0.34	0.58	0.53	0.57	0.54	0.64	0.67	0.62	0.55
	RMSE (t/ha)	0.90	0.92	0.92	0.97	0.78	0.80	0.76	0.80	0.73	0.67	0.76	0.80
	NRMSE	30.73 %	31.13 %	31.17 %	32.81 %	26.84 %	27.32 %	25.79 %	27.18 %	24.83 %	22.83 %	25.66 %	27.20 %
MT	R^2	0.49	0.46	0.56	0.39	0.52	0.52	0.53	0.59	0.78	0.66	0.60	0.62
	RMSE (t/ha)	0.86	0.86	0.79	0.92	0.81	0.80	0.81	0.75	0.63	0.69	0.80	0.73
	NRMSE	29.59 %	29.20 %	26.67 %	31.18 %	27.41 %	27.25 %	27.46 %	25.43 %	21.78 %	23.26 %	27.27 %	24.68 %
RMT	R^2	0.61	0.50	0.57	0.41	0.63	0.55	0.60	0.59	0.79	0.68	0.70	0.61
	RMSE (t/ha)	0.75	0.83	0.78	0.91	0.77	0.78	0.73	0.75	0.62	0.67	0.64	0.73
	NRMSE	25.04 %	28.19 %	26.51 %	30.86 %	25.69 %	26.46 %	24.94 %	25.41 %	21.55 %	22.60 %	21.86 %	24.85 %

Note: R: red-green-blue (RGB); M: multispectral (MS); T: thermal infrared (TIR); RM: RGB + MS; RT: RGB + TIR; MT: MS + TIR; RMT: RGB + MS + TIR; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor.

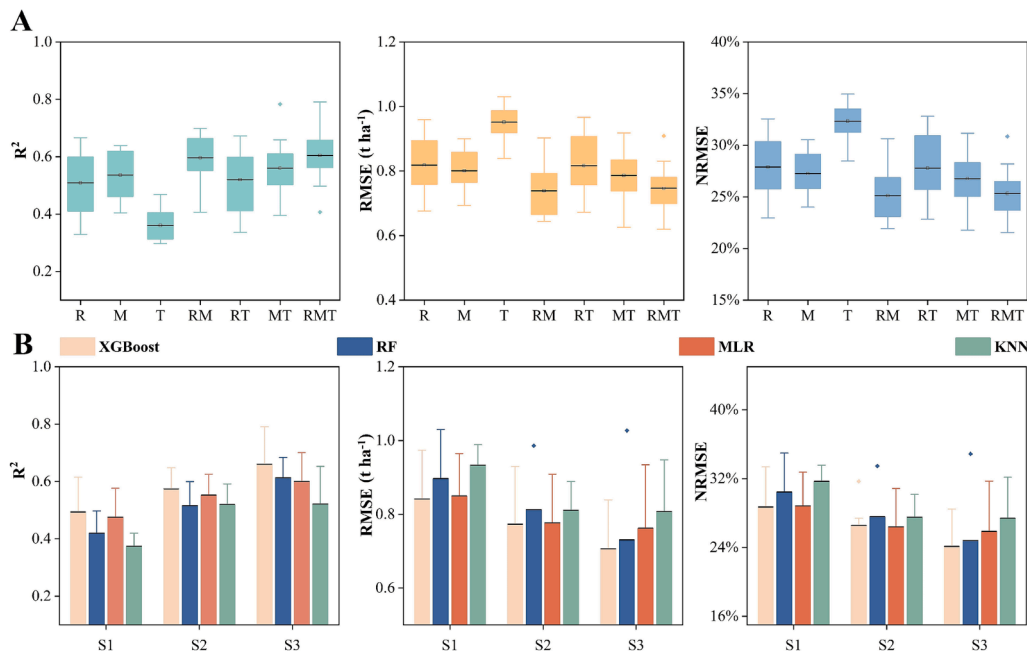


Fig. 9. The statistical distribution of the estimation accuracy for yield. A: the estimation accuracy of faba bean yield using different sensors; B: the estimation accuracy of faba bean yield using different models at three growth stages. R: red-green-blue (RGB); M: multispectral (MS); T: thermal infrared (TIR); RM: RGB + MS; RT: RGB + TIR; MT: MS + TIR; RMT: RGB + MS + TIR; S1: flowering stage; S2: podding stage; S3: grain filling stage; XGBoost: extreme gradient boosting; RF: random forest; MLR: multiple linear regression; KNN: k-nearest neighbor. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the fresh weight of crops highly depends on their water content. However, dry weight is primarily related to the accumulation of organic matter in the crop rather than water content. Therefore, the NRCT is an important feature in estimating fresh weight but it is determined to be an unimportant feature when estimating dry weight.

Although the MS sensor contained more spectral information, the RGB sensor exhibited a higher spatial resolution than the MS sensor (Meng et al., 2023). This can explain the higher AGB estimation accuracy of the RGB sensor compared to that of the MS sensor. Sensor fusion generally enhances estimation accuracy, which is consistent with

previous studies (Guo et al., 2022; Zhu et al., 2021). However, when estimating dry weight using the KNN and MLR, the RGB sensor outperformed the other sensors. This can be attributed to the fact that RM includes more spectral bands, thereby increasing the dimensionality of the feature space beyond the capabilities of MLR and KNN in handling high-dimensional information. Under cost constraints, the RGB sensor provides a better alternative for estimating the AGB of faba bean.

As faba bean plants grow, the CC gradually increases. Previous studies have demonstrated that as CC increases, VIs tend to saturate, resulting in a decrease in AGB estimation accuracy as the growth stage

progresses (Wang et al., 2022). For field managers, accurate monitoring of AGB during the early stages is vital because it can facilitate real-time field management and improve crop yield. The selection of appropriate algorithms significantly influences AGB estimation. Zhang et al. (2022) employed XGBoost, RF, MLR and univariate linear regression algorithms to develop a soybean LAI estimation model. They found that the XGBoost model exhibited the highest estimation accuracy, which is consistent with the findings of this study. The XGBoost algorithm penalizes the weights of leaf nodes to prevent overfitting, resulting in excellent optimization of gradient boosting.

4.3. Performance analysis of faba bean yield estimation

In this study, PH identified as an important variable in estimating yield at later stages, but it is not important at early stages. This may be because faba bean PH tends to be consistent in the early growth stage, but PH shows some variations in the later growth stage, so it is considered an important estimation variable in the podding and grain filling stage. The estimation potential for faba bean yield using RGB and MS sensors is relatively similar. Guo et al. (2022) estimated corn yield using both RGB and MS sensors and found that the MS sensor demonstrated slightly higher estimation accuracy, which is consistent with the results of this study. The advantage of the MS sensor depends on its sensitivity to vegetation in the near-infrared and infrared bands (Hatfield et al., 2015). However, the higher spatial resolution of the RGB sensor may have compensated for this disadvantage in this study. TIR sensors provide canopy temperature information, which is significantly correlated with factors such as leaf water content and crop transpiration (Maimaitijiang et al., 2020). However, in this study, the estimation accuracy of yield by TIR sensors is poor possibly owing to the sensitivity of canopy temperature to environmental temperature and water resource conditions. The estimation accuracy of faba bean yield could be effectively improved using fusion data from multiple sensors, which is consistent with previous studies (Fei et al., 2021b; Feng et al., 2020). The fusion of multiple sensors could combine the spectral characteristics of various sensors and contribute to the yield estimation.

Among the four algorithms, XGBoost achieved the highest estimation accuracy, which is consistent with the findings in Subsection 4.2,

However, the KNN algorithm exhibited the lowest accuracy in estimating faba bean yield. This may be because it depends significantly on training data and is more suitable for classification applications (Cheng et al., 2022). The growth stage significantly influences the estimation of crop physiological parameters using VIs (Liu et al., 2022). A higher estimation accuracy was achieved in the later growth stage than in the early growth stage, which is consistent with the results of the soybean yield estimation (Herrero-Huerta et al., 2020). However, this does not prove that the estimation in the later growth stage is more important. If the goal is to formulate the planting plan for next year and estimate economic value in advance based on yield estimation, then the later growth stage is a better choice. Conversely, if the goal is to improve cultivation measures promptly based on yield estimation, then the early growth stage is a better choice.

4.4. Estimation results and future prospects

As the results indicated, the XGBoost algorithm performed better than RF, MLR and KNN in estimating faba bean fresh weight, dry weight and yield. Figs. 10–13 presents the estimated PH and optimal estimation results of the AGB and yield obtained using the XGBoost algorithm during three growth stages. The fusion data result in improved estimation accuracy for both yield, fresh weight and dry weight. In future studies, sensor fusion will play an increasingly important role in phenotype estimation. Furthermore, advancements in ML and sensor technologies will enhance the accuracy of crop phenotypic traits estimation.

Although this study conducted experiments over two years and achieved encouraging results, there is still room for improvement. The experimental sites were relatively limited and did not consider environmental variables, which restricted large-scale applications. Additionally, the specific impact of drone flight altitude, image resolution, and data collection timing on the estimation accuracy needs further investigation. Previous studies (Liu et al., 2022a) and our study indicated that fewer features can also provide acceptable estimation accuracy. Thus, in future studies, we also need to explore the ability to estimate crop phenotypes using fewer features to reduce model complexity. Additionally, we should incorporate data from more time

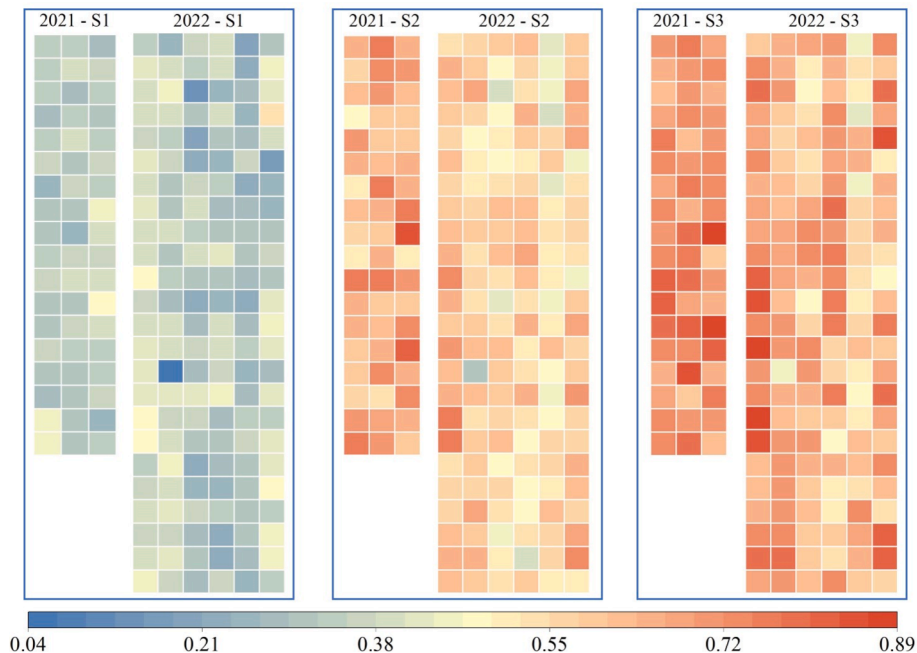


Fig. 10. Estimation map of faba bean plant height. The different color depict the value of plant height from low (blue) to high (red); S1: flowering stage; S2: podding stage; S3: grain filling stage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

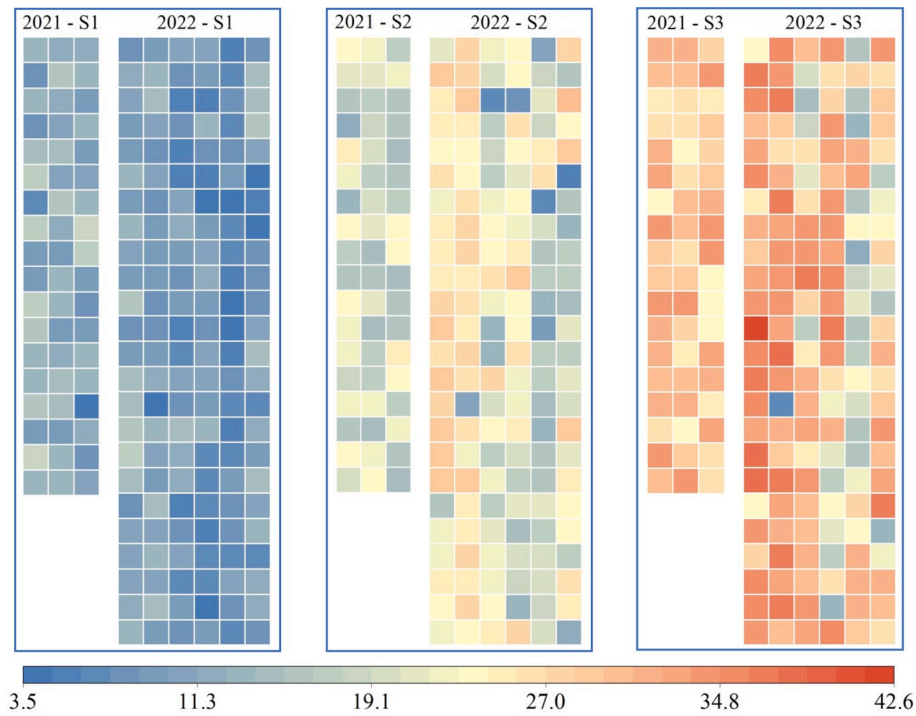


Fig. 11. Estimation map of faba bean fresh weight. The different color depict the value of fresh weight from low (blue) to high (red); S1: flowering stage; S2: podding stage; S3: grain filling stage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

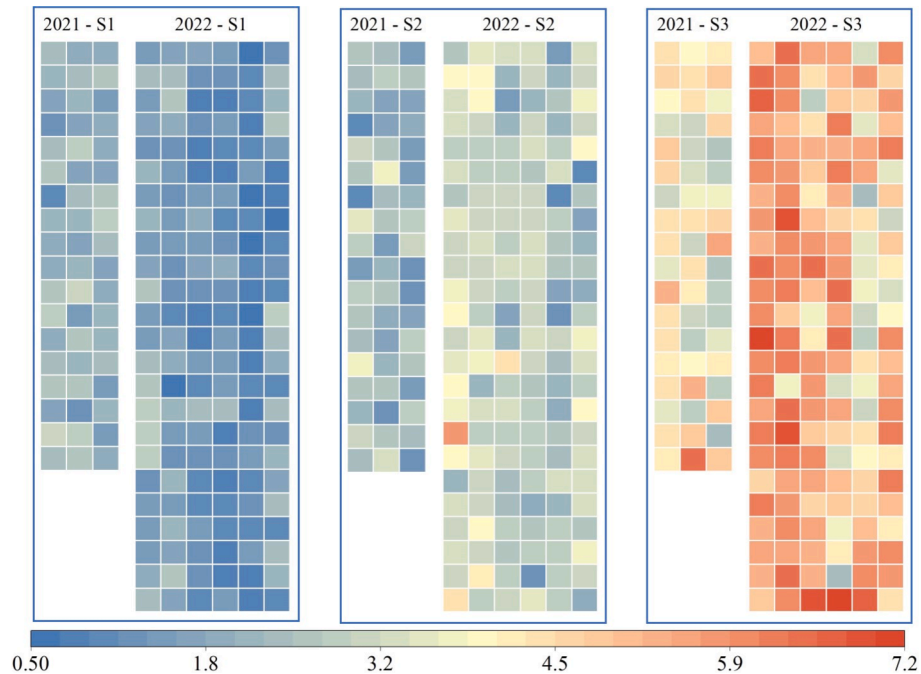


Fig. 12. Estimation map of faba bean dry weight. The different color depict the value of dry weight from low (blue) to high (red); S1: flowering stage; S2: podding stage; S3: grain filling stage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

points and locations, as well as consider factors such as climate and soil conditions, to enhance the model's generalizability. And further exploration of the impact of drone flight conditions on estimation accuracy. Hyperspectral sensors typically provide more spectral information. As researchers continue to explore the use of hyperspectral sensors for estimating crop phenotypes (Liu et al., 2023c), their accuracy has been proven in estimating potato biomass (Liu et al., 2024a; Liu et al., 2024b). Utilizing hyperspectral sensors to estimate faba bean biomass and yield

is expected to be an effective method for further improving estimation accuracy.

5. Conclusions

In this study, a two-year experiment was conducted from 2020 to 2022, during which UAV-based multimodal data were collected using RGB, MS and TIR sensors. Faba bean PH was estimated using the

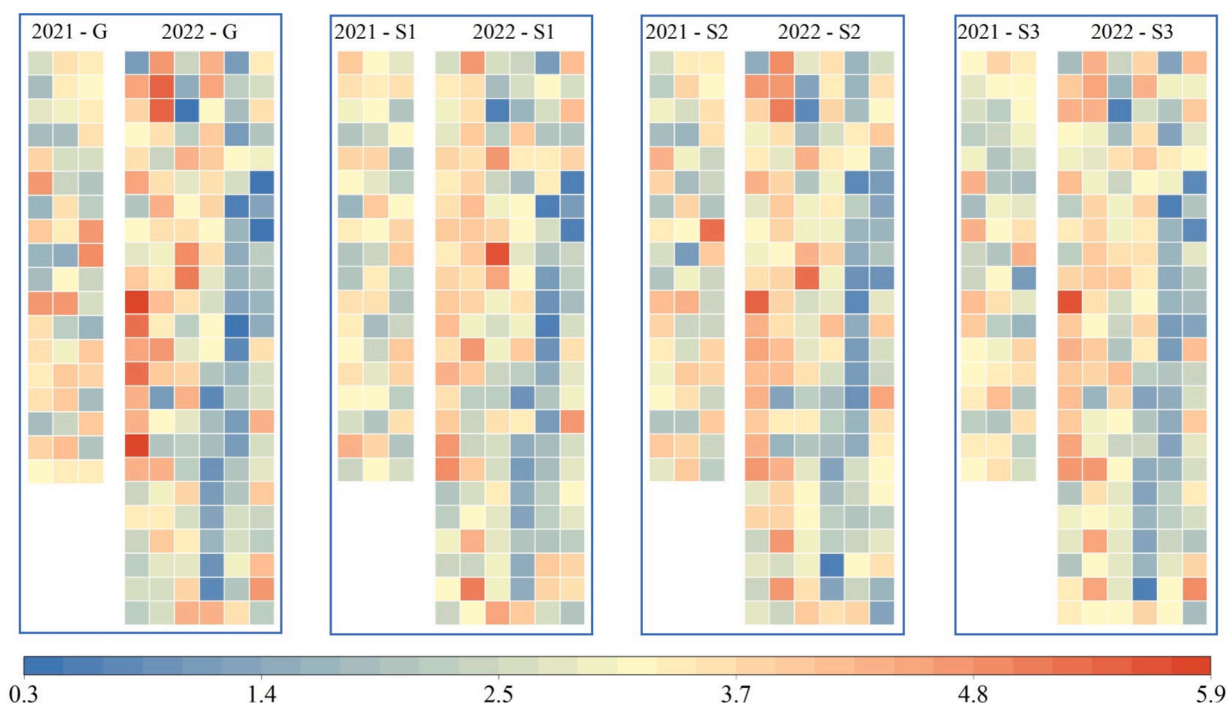


Fig. 13. Estimation map of faba bean yield. The different color depict the value of yield from low (blue) to high (red). G: ground measured; S1: flowering stage; S2: podding stage; S3: grain filling stage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

percentile method and phenotyping traits (fresh weight, dry weight and yield) by combining three sensors (RGB, MS and TIR) and four algorithms (XGBoost, RF, MLR and KNN). The conclusions are as follows.

- (1) Faba bean PH estimation using various percentiles consistently obtained reliable results, and PH_{max} was the preferred percentile.
- (2) The fusion of multiple sensors improved the accuracy of estimating the fresh weight, dry weight and yield of faba bean. However, the contribution of the TIR sensor to the estimation was relatively small, particularly during the dry weight estimation, the NRCT derived from the TIR sensor was identified as an unimportant variable.
- (3) In most cases involving single-sensor data or multi-sensor fusion data, the XGBoost algorithm demonstrated higher estimation accuracy than the RF, MLR and KNN algorithms. Therefore, XGBoost can be considered as a suitable choice for estimating the phenotypic traits of faba bean.

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CRedit authorship contribution statement

Yishan Ji: Writing – review & editing, Writing – original draft, Project administration, Investigation. **Zehao Liu:** Writing – original draft, Validation, Investigation. **Rong Liu:** Data curation. **Zhirui Wang:** Data curation. **Xuxiao Zong:** Supervision, Resources, Funding acquisition. **Tao Yang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109584>.

Data availability

Data will be made available on request.

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